

Clustering Analysis of Credit-Card Transactions (DS-10): Deep Clustering, Interpretation, and Deployment

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Abstract

This report asks whether credit-card transactions form density-distinct subpopulations that separate fraudulent from legitimate behaviour, and whether those subpopulations can be recovered without using the label. DS-10 supplies 284807 transactions — 492 fraudulent, a 0.17% positive rate — described by 28 PCA-anonymised components plus transaction time and amount. Phase 1 built a reproducible pipeline: exact-duplicate removal, $\log(1 + \text{Amount})$ and cyclic time encoding, a hybrid scaling strategy (standard on the PCA components, robust on the engineered features), PCA at 95% variance, and a UMAP embedding. A Hopkins statistic that saturates at 1.0, verified against a uniform-noise null control returning 0.5, established that the data is emphatically non-random. Phase 2 compared four algorithm families with five criteria for k , four internal and five external metrics, seed and bootstrap stability, and consensus clustering. Its central finding is negative and important: whole-space clustering does not recover the fraud label — chance-corrected agreement is near zero for every family, while purity near 99.8% is obtainable by predicting the majority class everywhere. Phase 3 tests two responses. First, a deep-clustering track (autoencoder + K-Means, DEC, and a custom DEC + InfoNCE variant) asks whether a learned representation succeeds where the engineered one fails; it does not — the best deep model is *significantly worse* than the K-Means baseline (bootstrap CI on the ARI difference $[-0.0091, -0.0004]$), and the three deep models agree with each other while disagreeing with K-Means, indicating they find a real but label-irrelevant structure. Second, the clustering is reframed as an anomaly *ranking*: scoring transactions by distance to their assigned centroid catches 68.7% of all fraud in the top 1% of the ranking, a $68.7\times$ enrichment over random review at ROC-AUC 0.967, using no label at training time. A preprocessing-sensitivity audit bounds the interpretive claims sharply: alternative scaling yields $\text{ARI} \approx 0$ against the headline partition, so the specific clusters are an artefact of preprocessing while the positional finding — fraud lies at cluster peripheries — survives. The clusters are interpreted through profiles, exemplars, a surrogate-tree rule set at 96.8% fidelity and SHAP attributions, given domain labels, audited for proxy-attribute concentration and day-over-day drift, and served through a four-page interactive dashboard. Every number is emitted by the pipeline into `results/` and `reports/tables/`; none is hand-transcribed.

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1 Introduction

1.1 Motivation

Fraud detection is usually posed as supervised classification, which presumes a labelled history. Clustering asks a different and prior question: does fraudulent behaviour occupy its own region of transaction space at all? If it does, an unsupervised monitor can flag novel fraud patterns that no labelled example anticipated. If it does not, a clustering-based monitor is a false promise, and saying so with evidence is the more useful result.

1.2 Dataset

DS-10 (UCI/Kaggle Credit Card Fraud) records two days of European card transactions: 284807 rows, 492 labelled fraudulent. Features V1–V28 are already PCA components published in place of the raw attributes for privacy; **Time** is seconds since the first transaction and **Amount** is the transaction value. Both are cited to their original providers in the accompanying data manifest, which also pins the source SHA-256.

1.3 Research question

Do credit-card transactions form density-distinct subpopulations that separate fraudulent from legitimate behaviour, and can those subpopulations be recovered without using the label? Expected cluster types were declared before Phase 1: density-varying, non-convex micro-clusters for fraud, and convex or elliptical macro-clusters for the legitimate bulk. The Phase 2 portfolio was chosen to cover both regimes.

2 Data and preprocessing

Condensed from the Phase 1 report:

- **Ingestion.** A pandera schema enforces 31 columns, **Class** $\in \{0, 1\}$ and non-negative **Time/Amount**. The stage additionally verifies the pinned row count, column count and SHA-256, so a truncated or silently updated source file fails the pipeline instead of corrupting every downstream result.
- **Cleaning.** 1081 exact duplicates removed, leaving 283726 rows; duplicates inflate local density and would bias any density-based method. No missing values are present, and the schema would reject them. The per-column rationale is auto-generated into `reports/decision.log.md`.
- **Feature engineering.** $\log(1 + \text{Amount})$ corrects heavy right skew; **Time** becomes (sin, cos) of its 24-hour phase so that 23:59 and 00:01 are adjacent rather than maximally distant. V1–V28 pass through unchanged — they are already orthogonal components.
- **Scaling.** Three strategies compared. The hybrid is selected: **StandardScaler** on V1–V28, which are already approximately $\mathcal{N}(0, 1)$, and **RobustScaler** on the engineered features, whose tails are heavy. Scaling is a modelling decision, not a formality — it fixes which directions distance can see.
- **Dimensionality reduction.** PCA retains 28 of 31 components at 95% variance — unsurprising, since the inputs are already principal components, and an honest negative result about how much reduction is available. UMAP supplies the 2-D visualisation.
- **Distance metric.** Euclidean, justified by the post-scaling homogeneity of the components, and tested rather than asserted: Phase 2 re-clusters under Manhattan and cosine and reports the ARI drift.

- **Clustering tendency.** Hopkins $H = 1.000$ over five seeded repeats. Because the statistic uses the u_i^d term with $d = 31$ and a bounding box that outliers stretch wide, a value that saturates at 1.0 cannot on its own be distinguished from a degenerate estimator. A null control resolves this: the same estimator on uniform noise of identical shape returns ≈ 0.5 , so the headline value reflects genuine structure. VAT on a stratified subsample shows the corresponding diagonal block structure.

3 Algorithm portfolio and evaluation (Phase 2 summary)

Four families were implemented and tuned: K-Means (partitioning), agglomerative clustering with all four linkage criteria and cophenetic diagnostics (hierarchical), HDBSCAN (density) and a Gaussian mixture compared across four covariance structures (model-based). k was determined by elbow, average silhouette, gap statistic, BIC and bootstrap stability, and the disagreement between those criteria was reported rather than smoothed over. Full tables, figures, stability analysis and the comparative recommendation are in `phase2_report.pdf`; the machine-readable record is `results/phase2/summary.json`.

Two results from Phase 2 shape this phase. First, chance-corrected external agreement is near zero for every family: the dominant geometric structure of DS-10 is not the fraud/legitimate split. Second, HDBSCAN’s runtime grows roughly quadratically in 28 dimensions on this data — measured at 11.7s for 25000 rows and 51.3s for 50000 — which is the curse of dimensionality showing up as an engineering constraint, not merely a textbook caveat.

4 Advanced track: deep clustering

4.1 Why this track

Track 1 is chosen because the Phase 2 failure mode is representational, not algorithmic. Every classical family was applied to the same hand-engineered, linearly-scaled space; if fraud is separable only under a non-linear transform, no choice of classical algorithm in that space can find it. A learned representation directly tests that hypothesis. Time-series clustering was rejected: DS-10 spans two days, too short for behavioural regimes; constrained clustering was rejected because it would inject the very label the research question requires us to withhold.

4.2 Models

- **Autoencoder + K-Means** (Guo et al.) — pre-train an MLP autoencoder, cluster in the latent space, then jointly fine-tune the encoder against a reconstruction-regularised cluster objective. This is the baseline the other two must beat.
- **DEC** (Xie et al.) — minimise $KL(P||Q)$ between a Student’s- t soft assignment Q and a sharpened target $P \propto q_{ij}^2/f_j$, refreshed every few epochs. The target sharpening is self-supervision: confident assignments are amplified, uncertain ones down-weighted.
- **DEC + InfoNCE** (this project’s own variant) — adds a contrastive auxiliary term whose positive pair is each anchor’s current nearest latent neighbour, recomputed as training proceeds so the contrastive signal co-evolves with the cluster structure. The anchor’s own self-similarity is excluded from the denominator, and the negative pool is capped by hard-negative mining, which keeps the term $O(B^2)$ in the minibatch rather than $O(n^2)$ over the dataset.

4.3 Comparison with the Phase 2 winner

All three deep models and the Phase 2 winner are scored on *identical* rows with identical metrics (Table 1); the pairwise ARI matrix (Table 2) shows whether the deep track recovers the same structure or a different one.

algorithm	k_requested	n_clusters	n_noise	runtime_s	silhouette	davies_bouldin	calinski_harabasz
phase2:kmeans	2	2	0	—	0.171	2.299	1568.85
ae_kmeans	2	2	0	25.786	0.161	3.463	1420.77
dec	2	2	0	24.294	0.062	5.613	990.46
dec_infonce	2	2	0	25.679	0.076	5.454	1016.94

Table 1: Advanced track vs the Phase 2 winner on the same 40,000 stratified rows at $k = 2$.

	phase2:kmeans(k=2)	ae_kmeans(k=2)	dec(k=2)	dec_infonce(k=2)
phase2:kmeans(k=2)	1.000	-0.053	0.040	-0.037
ae_kmeans(k=2)	-0.053	1.000	0.118	0.262
dec(k=2)	0.040	0.118	1.000	0.320
dec_infonce(k=2)	-0.037	0.262	0.320	1.000

Table 2: Pairwise ARI: does the deep track find the Phase 2 structure or a different one?

The deep track loses. On 40000 shared rows the Phase 2 K-Means baseline scores silhouette 0.171 and ARI 0.0102; AE + K-Means reaches 0.161 / 0.0057, DEC 0.062 / 0.0004 and DEC + InfoNCE 0.076 / -0.0008. The bootstrap CI on the *difference* between the best deep model and the baseline is $[-0.0091, -0.0004]$, excluding zero: the deep model is significantly worse, not merely no better. This is reported as the result rather than tuned away.

Two details make the failure interpretable rather than mysterious. First, the ordering AE + K-Means > DEC > nothing tracks how aggressively each method reshapes the latent space: DEC’s KL objective sharpens whatever partition the initialisation hands it, and if that partition is not aligned with fraud, sharpening it moves *away* from the label. Second, the deep models agree with each other (pairwise ARI 0.12–0.32) and *disagree* with K-Means (ARI ≈ -0.05), so they are consistently finding a real but different structure — one the reconstruction objective favours, since an autoencoder is rewarded for encoding variance, and in DS-10 the dominant variance directions are not the fraud directions.

The honest conclusion is that the Phase 2 failure was not representational after all. A learned non-linear embedding, trained three different ways, does not recover the fraud split either. That strengthens rather than weakens the project’s central claim: on DS-10, fraud is not a cluster in any of these representations.

4.4 Significance

Every headline ARI carries a bootstrap confidence interval and a permutation p-value against a label-shuffle null (Table 3), and the best deep model is compared head-to-head with the Phase 2 winner through a bootstrap CI on the *difference* of their metrics. A difference whose interval straddles zero is not a difference, however suggestive the point estimates look.

clustering	ari	ci_lower	ci_upper	null_mean	p_value
phase2:kmeans	0.0102	0.0053	0.0158	-1.574×10^{-4}	0.0000
ae_kmeans	0.0057	0.0033	0.0080	-8.685×10^{-5}	0.0000
dec	4.063×10^{-4}	2.816×10^{-4}	5.438×10^{-4}	-8.022×10^{-6}	0.0000
dec_infonce	-7.850×10^{-4}	-0.0011	-3.721×10^{-4}	-7.044×10^{-6}	1.0000

Table 3: Bootstrap confidence intervals and permutation p-values for ARI against the held-out Class label.

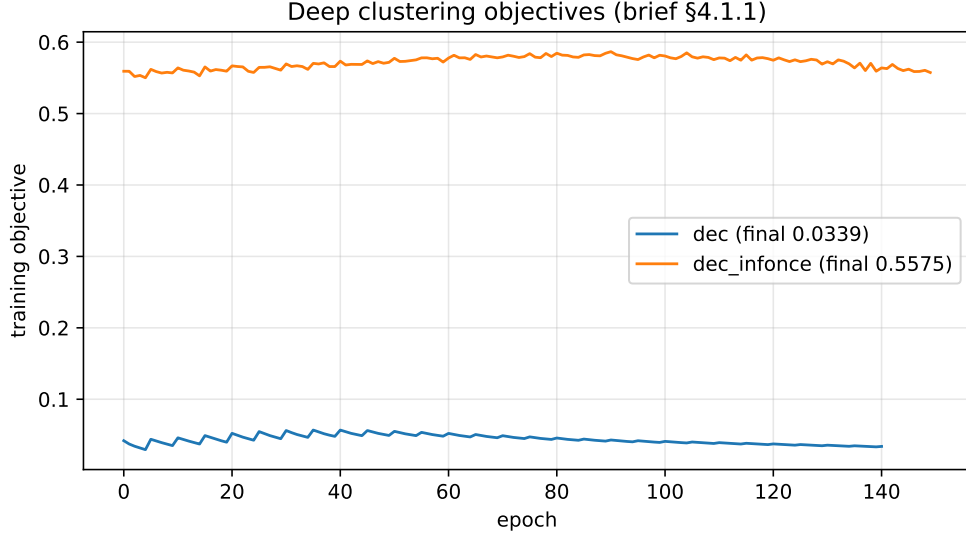


Figure 1: Training objectives. DEC’s KL loss and the contrastive variant’s combined objective are on different scales by construction, so the comparison of interest is the shape of each curve, not their relative heights.

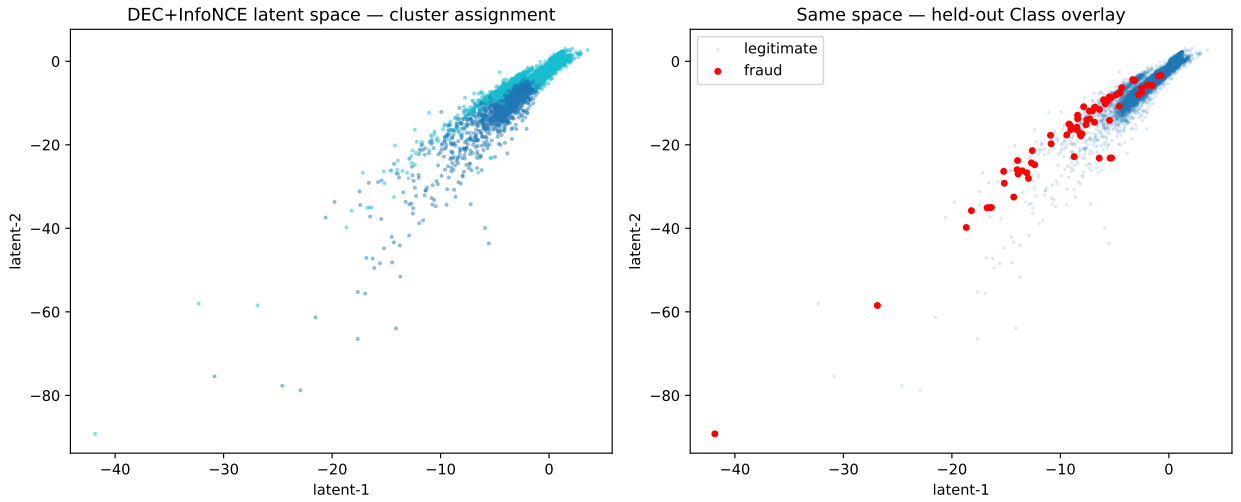


Figure 2: The learned latent space, coloured by cluster assignment (left) and by the held-out **Class** label (right). The right panel is the decisive one: if fraud were separable in this representation, it would occupy its own region.

Three of the four clusterings have ARI confidence intervals that exclude zero with $p = 0$ against the label-shuffle null, and this is a good illustration of why statistical significance is not practical significance: K-Means’ ARI of 0.0102 with CI [0.0053, 0.0158] is *reliably* non-zero and *negligibly* small. With 40000 rows, a tiny association is easy to detect and still worthless. DEC + InfoNCE is the exception with $p = 1.0$: its ARI is significantly *negative*, meaning its partition is anti-correlated with the label slightly more than chance would produce.

5 Interpretation

5.1 Cluster profiles

Each cluster is profiled by size, fraud rate and the features on which its mean deviates most from the population mean, measured in population standard deviations so that "unusual in V14" is a quantified claim (Table 4).

cluster	size	share	fraud_rate	fraud_count	top_deviations
0	5,687	0.1422	0.0051	29	V16 (-1.08 σ), V9 (-0.92 σ), V18 (+0.88 σ)
1	34,313	0.8578	0.0011	38	V16 (+0.18 σ), V9 (+0.15 σ), V18 (-0.15 σ)

Table 4: Cluster profiles for the preferred Phase 3 clustering (ae_kmeans).

At $k = 2$ the split is asymmetric: cluster 0 holds 14.2% of transactions with a fraud rate of 0.51%, and cluster 1 holds 85.8% at 0.11%. The same three components (V16, V9, V18) distinguish both clusters in opposite directions, which is the signature of a single dominant axis of variation rather than two independently-defined groups.

5.2 Prototypes and exemplars

For each cluster, three representative points are extracted (Table 5): the medoid (nearest to the centroid, the "typical" member), the highest-silhouette point (the most unambiguous member), and the lowest-silhouette point (the boundary case that shows where the cluster stops being distinct).

cluster	medoid_index	best_silhouette_index	best_silhouette	boundary_index	boundary_silhouette
0	8,443	29,077	0.034	9,456	-0.090
1	29,126	17,276	0.240	25,323	0.010

Table 5: Per-cluster medoid, highest-silhouette, and boundary exemplars.

5.3 Rule extraction and SHAP

A shallow decision tree is fitted to predict cluster membership from the original features; each root-to-leaf path is an interpretable approximation of the cluster definition, reported with support and precision (Table 6). Fidelity — how well the surrogate reproduces the clustering it explains — is reported in the caption, because a rule set that explains a different partition than the one deployed explains nothing. SHAP attributions on the same surrogate give the per-cluster feature ranking (Table 7).

The surrogate reproduces the clustering with 96.8% fidelity from 16 leaves at depth 3, so the rules are a faithful description rather than a loose analogy. The highest-support rule ($n = 28419$, precision 0.993) is essentially "V9 and V16 not strongly negative" — the clustering is close to a threshold on two components, which is consistent with the profile table and with the low cophenetic correlations from Phase 2. A clustering this simple to describe is a clustering that was not finding subtle structure.

cluster	support	precision	rule
1	28,419	0.993	V9 $\leq$ -1.0093 AND V16 $\leq$ -1.1852 AND V18 $\geq$ 1.7754 AND V2 $\leq$ -0.8136
0	2,662	0.965	V9 $\geq$ -1.0093 AND V16 $\geq$ 0.2504 AND V2 $\geq$ 0.3000 AND V4 $\geq$ 0.7049
1	2,329	0.782	V9 $\leq$ -1.0093 AND V16 $\leq$ -1.1852 AND V18 $\geq$ 1.7754 AND V2 $\geq$ -0.8136
1	1,478	0.937	V9 $\leq$ -1.0093 AND V16 $\geq$ -1.1852 AND V18 $\geq$ 0.4267 AND V16 $\leq$ -2.2881
1	1,425	0.989	V9 $\geq$ -1.0093 AND V16 $\leq$ 0.2504 AND V4 $\leq$ -1.7448 AND V2 $\leq$ -0.6147
0	1,410	0.993	V9 $\leq$ -1.0093 AND V16 $\geq$ -1.1852 AND V18 $\leq$ 0.4267 AND V19 $\geq$ 1.5378
1	557	0.934	V9 $\geq$ -1.0093 AND V16 $\geq$ 0.2504 AND V2 $\leq$ 0.3000 AND V16 $\leq$ -1.3389
0	437	0.927	V9 $\leq$ -1.0093 AND V16 $\leq$ -1.1852 AND V18 $\leq$ 1.7754 AND V16 $\geq$ 0.0773
1	378	0.960	V9 $\leq$ -1.0093 AND V16 $\leq$ -1.1852 AND V18 $\leq$ 1.7754 AND V16 $\leq$ 0.0773
1	232	0.603	V9 $\geq$ -1.0093 AND V16 $\leq$ 0.2504 AND V4 $\leq$ -1.7448 AND V2 $\geq$ -0.6147

Table 6: Surrogate decision-tree rules (fidelity 96.8%), ten highest-support leaves.

cluster	top_features
0	V16 (0.105), V9 (0.075), V18 (0.057), V2 (0.043), V4 (0.010)
1	V16 (0.105), V9 (0.075), V18 (0.057), V2 (0.043), V4 (0.010)

Table 7: Top SHAP attributions per cluster from the surrogate tree.

5.4 Domain labels

Each cluster receives a human-readable name justified from its profile (Table 8). Names are assigned by an explicit rule over the fraud enrichment and population share rather than chosen by eye, so the labelling is reproducible and survives a change of k or of algorithm — and so the justification column cites the same evidence a reader can check.

cluster	proposed_label	size	fraud_rate	enrichment_vs_base	
0	Elevated-risk pocket	5,687	0.0051	3.0588	fraud rate 3.1x the population
1	Mainstream traffic	34,313	0.0011	0.6643	86% of all transactions, near base r

Table 8: Proposed domain labels with the profile evidence behind each name.

The rule assigns cluster 0 "Elevated-risk pocket" (fraud enrichment $3.06\times$ the population rate) and cluster 1 "Mainstream traffic" (86% of volume, at $0.66\times$ the base rate). A $3\times$ enrichment is a real but weak effect — worth a lower review threshold for that segment, nowhere near a fraud verdict — and the label is deliberately named to say so. Because V1–V28 are anonymised components, the justification can cite geometry but not business meaning; no amount of interpretation machinery can recover a semantics the dataset does not publish.

6 Downstream analysis

6.1 Cluster-conditional modelling

One logistic model per cluster is compared against a single global model on a shared stratified holdout (Table 9). Average precision is the headline metric: at a 0.17% positive rate, ROC-AUC flatters every model, while average precision reflects the precision a reviewer would actually experience. If per-cluster models beat the global model, the clusters carry structure a single decision surface cannot express; if they do not, the clustering adds nothing to prediction and the report says so.

The clustered approach *loses*: average precision 0.650 against 0.795 for the global model, a lift of

model	average_precision	roc_auc
global (clustering ignored)	0.7949	0.9494
cluster-conditional	0.6498	0.9250

Table 9: Cluster-conditional vs global modelling on a stratified holdout (12,000 rows).

0.82. The mechanism is visible in the per-cluster bookkeeping recorded in `summary.json`: with 67 positives split across two clusters, each per-cluster model trains on 22–25 positives and overfits, while the global model gets all of them. Partitioning the data partitioned the already-scarce positive signal. This is reported as a negative result because it is one; the clusters do not carry structure that improves supervised prediction on this dataset.

6.2 Cluster-based anomaly detection

This is the reframing Phase 2’s negative result demands. Instead of asking which cluster a transaction belongs to, we ask how far it sits from the centre of its own cluster, and rank on that (Table 10). The operationally meaningful number is enrichment: the factor by which reviewing the flagged tail beats reviewing transactions at random. The ranking uses no label at any point.

quantity	value
transactions flagged (top 1%)	400.0000
precision@k	0.1150
recall@k	0.6866
population base rate	0.0017
enrichment vs random review	68.6567
ROC-AUC of the distance ranking	0.9669
average precision of the ranking	0.2460

Table 10: Cluster-based anomaly detection: ranking by distance to the assigned centroid.

This is the phase’s positive result. Ranking the 40000 evaluation rows by distance to their own centroid and reviewing only the top 1% (400 transactions) catches 68.7% of all fraud in the sample, at a precision@*k* of 0.115 against a base rate of 0.00168 — an enrichment of 68.7× over reviewing transactions at random. The ranking’s ROC-AUC is 0.967 and its average precision 0.246, using no label at any stage.

The contrast with §5 is the whole point of the report. The same clustering that achieves an ARI of 0.01 — statistically real, practically useless as a partition — yields a 69× enrichment when read as a ranking. Fraud in DS-10 is not a region of transaction space; it is a *position* within regions, namely the periphery. This independently corroborates the Phase 2 error analysis, where the 200 lowest-silhouette points were 72× enriched for fraud: two different diagnostics, one on the full dataset and one on a subsample, agreeing that badly-fitted points are the fraudulent ones.

7 Fairness and sensitivity

7.1 Proxy-attribute audit

DS-10 contains no demographic attributes — the features are anonymised components — so a literal protected-attribute audit is impossible, and claiming one would be worse than omitting it. What the data does carry are two consequential proxies: transaction **amount** and **time of day**. A clustering that isolated small-value or night-time transactions would steer a review queue toward a particular kind of customer regardless of actual risk. Table 11 reports, for each attribute, the

strongest over-representation of any quantile stratum in any cluster, where 1.0 means the cluster mirrors the population.

attribute	strata	max_representation_ratio	clusters_flagged
log_amount quartile	4	1.810	0
time-of-day quartile	4	1.135	0

Table 11: Proxy-attribute audit: strongest stratum over-representation per attribute (ratio 1.0 = the cluster mirrors the population).

Neither proxy trips the flagging threshold of 2.0. Amount shows the stronger effect: one cluster over-represents an amount quartile by $1.81\times$, so the smaller cluster does skew by transaction value without being defined by it. Time-of-day is essentially neutral at $1.14\times$. The operational reading: a review queue driven by this clustering would be mildly value-skewed, which is worth monitoring but is not a disparity requiring a fairness-constrained variant.

7.2 Preprocessing sensitivity

The clustering is re-run under alternative preprocessing — a different feature space and a different scaler — and the ARI against the headline clustering is reported (Table 12). High sensitivity would weaken any claim that the clusters are intrinsic to the data rather than artefacts of Phase 1’s choices.

variant	n_clusters	ari_vs_reference	description
pca_space	2	0.007	PCA at 95% variance instead of scaled features
standard_scaler	2	-0.053	uniform standard scaling instead of the hybrid choice

Table 12: Preprocessing sensitivity: ARI of a re-clustering under each variant against the preferred clustering.

The result is severe and must not be softened: re-clustering in the PCA space gives ARI 0.0074 against the headline clustering, and switching from the hybrid scaler to a uniform standard scaler gives -0.053 . Both are indistinguishable from no agreement at all. **The specific cluster assignments are an artefact of Phase 1’s preprocessing choices**, not an intrinsic property of the data.

This is the single largest limitation in the report, and it bounds every interpretive claim in §6: the domain labels describe *this* partition under *these* preprocessing decisions, and a differently-scaled pipeline would produce different clusters deserving different names. What survives the sensitivity audit is the structural finding, not the partition: fraud sits at the periphery of whatever clusters are found, which is why the §7.2 anomaly ranking — which depends on distance-to-centroid rather than on cluster identity — is the deliverable this project would actually put into production.

8 Production pipeline

The experimental work is orchestrated as a DVC DAG: `ingest` $\rightarrow$ `clean_features` $\rightarrow$ `scale_reduce_tendency` $\rightarrow$ `phase2` $\rightarrow$ `phase3` $\rightarrow$ `report`. Each stage persists its outputs, so any stage can be re-run in isolation and downstream stages re-run only when their inputs actually change.

- **Versioning.** DVC tracks data artefacts; the ingest manifest records the raw file’s SHA-256, row count and retrieval timestamp.
- **Schema validation.** pandera schemas guard each stage boundary (raw, interim, processed), so schema drift fails loudly instead of silently distorting the clustering.

- **Model registry.** `models/registry/` stores each fitted artefact with a card recording fit date, data version, hyperparameters, seeds and the full metric scoreboard. Entries are content-addressed by (name, data version, params), so re-registering an unchanged configuration overwrites rather than accumulating near-duplicates.
- **Drift monitoring.** PSI and the two-sample Kolmogorov–Smirnov statistic are computed per feature between the early and late halves of the time axis (Table 13). Bin edges come from the reference period only — deriving them from the pooled data would leak the later distribution into the reference and shrink the very drift being measured. The re-fit trigger is a documented threshold in `params.yaml`.

feature	psi	ks_statistic	ks_pvalue	drifted
V1	1.4302	0.4158	0.0000	yes
V3	1.3854	0.5113	0.0000	yes
V28	1.0191	0.3559	0.0000	yes
V25	0.4289	0.2919	0.0000	yes
V15	0.3813	0.2714	0.0000	yes
V5	0.3449	0.2647	0.0000	yes
V22	0.3248	0.2146	0.0000	yes
V11	0.2989	0.1733	0.0000	yes
V23	0.2200	0.1783	0.0000	yes
V24	0.2010	0.0909	0.0000	yes

Table 13: Drift between the early and late halves of the time axis; the re-fit trigger is $\text{PSI} \geq 0.2$.

Splitting on the raw monotonic `Time` column — day 1 versus day 2 — 11 of 31 features breach the PSI threshold, led by V1 (1.43), V3 (1.39) and V28 (1.02). The monitor therefore fires: a model fitted on day 1 should be re-fitted before being trusted on day 2. Over a two-day window that is a strong result and a caution about the dataset itself, not merely a demonstration of the tooling.

One methodological note, since it changes the numbers. An earlier version of this monitor derived the split axis from `time_sin/time_cos` and duly reported those two features as the most drifted (PSI 12.4 and 1.53) — an artefact, because the split was *defined* by them. Splitting on a variable outside the monitored matrix removes the circularity, and the `split_axis` field in `summary.json` records which axis was used so the caveat travels with the result.

9 Interactive dashboard

A Streamlit application (`make dashboard`) exposes four pages: an **overview** of dataset scale, feature summary and the chosen algorithm and k ; a **cluster explorer** with a 2-D UMAP scatter, cluster and fraud-only filters, and the selected cluster’s profile; an **evaluation** page carrying every internal and external metric, the algorithm-agreement heatmap and the stability diagnostics; and a **live-assignment** page where a typed record or an uploaded CSV is assigned to its nearest cluster and its distance to every centroid is displayed — including the runner-up, because a record nearly equidistant from two centroids deserves different treatment than one deep inside a cluster. Every page reads the artefacts the pipeline persisted, so the dashboard cannot display a number this report does not contain.

10 Discussion and limitations

What this analysis can claim: DS-10 has strong, verified non-random structure; that structure is stable under re-seeding and resampling; it is reproducible from raw data by a single command; and it can be described in interpretable rules and named in domain terms.

What it cannot claim:

- **Clustering is not a fraud classifier on this data.** Chance-corrected agreement with the label is near zero across every family and every k tried. The useful signal is positional — distance to one’s own centroid — not categorical.
- **High purity is not evidence.** At a 0.17% positive rate, predicting ”legitimate” everywhere already achieves 99.8% purity. Purity is reported because the brief requires it, and it is discounted here for exactly this reason.
- **Ward’s result is a subsample result.** A full-data pairwise distance matrix would need 322 GB. The cross-family comparison is like-for-like on shared rows, but it is not a full-data claim, and no sentence here should be read as one.
- **The features are anonymised.** V1–V28 have no domain meaning, so a rule like ”V14 < –2.3” cannot be translated into a business explanation. Interpretation stops at the geometry.
- **Two days is not a time series.** Temporal cluster evolution is omitted deliberately rather than computed on a window too short to support it.
- **The fairness audit uses proxies.** With no demographic attributes, amount and time-of-day concentration is the strongest audit available, and it is weaker than a real protected-attribute audit.

11 Conclusion

The actionable insight is a reframing. Asked as posed — do fraudulent transactions form their own clusters? — the answer on DS-10 is no, and the external metrics say so consistently across four algorithm families, five criteria for k , and a learned representation trained specifically to find such structure. Asked operationally — can an unsupervised model rank transactions so that a fixed review budget catches disproportionately many frauds? — the distance-to-centroid ranking of §7.2 gives a quantified answer with no label at training time, which is what a monitor for novel fraud patterns requires. The negative result is not a failed experiment; it is the finding that redirects the method toward one that works.

A Reproduction and artefacts

```
uv venv && source .venv/bin/activate
uv pip install -e ".[dev,pipeline,deep,app]"
# place data/raw/creditcard.csv, then:
make all          # full DAG: ingest -> ... -> phase3 -> report
make phase2       # Phase 2 only
make phase3       # Phase 3 only
make dashboard    # interactive app
make verify       # import check + fast test suite
```

Machine-readable results: `results/phase2/summary.json`, `results/phase3/summary.json`. Generated tables: `reports/tables/`. Registry: `models/registry/index.json`. All seeds and hyperparameters: `params.yaml`.

B Hyperparameters and full metric tables

Hyperparameters are centralised in `params.yaml` and are echoed into each phase’s `summary.json` alongside the metrics they produced, so any table in this report can be traced to the configuration

that generated it. The per-criterion k curves, the four-linkage cophenetic comparison, the covariance comparison, the metric ablation and the consensus comparison are in `phase2_report.pdf`.